



Modeling knowledge proficiency using multi-hierarchical capsule graph neural network

Zeyu He^{1,2,3} · Wang Li¹ · Yonghong Yan^{1,4,5}

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Abstract

Knowledge Tracking (KT) predicts student performance by modeling the mastery of knowledge components in past interactions. Although the latest research on KT is excellent to model behavior changes, the increasingly complex knowledge components and semantic relationships of new exercises have made courses sparse and difficult to adequately diagnose knowledge mastery. Hence, it is not enough to satisfy only the personalized learning. This paper proposes a graph-based method with multi-hierarchical network (Caps-HAGKT) for tracking capsule knowledge, which integrates rich structural information of the knowledge space, including multiple skills, prerequisites, attribution, and conceptual meta-paths. Firstly, we use the multi-hierarchical knowledge capsule network to track a variety of high-level knowledge concepts, and then extract the latent knowledge structure between levels. In addition, we adopt a new graph network with an external memory matrix to model the relationship between concept capsules at the same level and update concept states. In previous works, it was difficult to use complex graph information without any expert pre-definition. In Caps-HAGKT, the semantic attention of knowledge capsules is used to merge different semantics in knowledge according to their importance. Extensive experiments on some real-world datasets show that the prediction performance of Caps-HAGKT is better than that of the existing KT method (AUC is increased by 3.05% on an average). The visualization of both latent graphs and case studies indicates the excellent interpretability of predictions.

Keywords Knowledge Tracing · Graph neural network · Capsule network · Learning sciences

1 Introduction

In the cognitive process, the human intelligence can quickly

focus on relevant entities, analyze facts, infer evolution, and support decision-making [1, 2]. A powerful cognition can be understood as the ability to locate knowledge entities and track their evolution, which enable humans to continuously master unknown knowledge and personalize the learning experience, thereby inspiring the desire for knowledge. Knowledge tracking is an elemental research which can effectively improve the participation rate and success rates of educational institutions. It is widely used in online open courses, intelligent tutoring systems, educational games, etc. Knowledge tracking aims to assess the levels of knowledge of students by observing a series of learning activities. It helps to innovate human teaching management techniques to improve learning efficiency and achieve personalized learning. For example, the teaching system can perceive the learning needs of students for providing knowledge remediation and recommending personalized courses [3]. Students recognize their own levels and devote more energy to knowledge weaknesses. Educational experts optimize the design of the teaching contents and methods

✉ Wang Li
wangli@hcl.ioa.ac.cn

Zeyu He
hezeyu@hcl.ioa.ac.cn

- ¹ Key Laboratory of Speech Acoustics and Content Understanding, Institute of Acoustics, Beijing, China
- ² Institute of Information Engineering, Chinese Academy of Sciences, Beijing, China
- ³ School of Cyber Security, University of Chinese Academy of Sciences, Beijing, China
- ⁴ Xinjiang Laboratory of Minority Speech and Language Information Processing, Xinjiang Technical Institute of Physics and Chemistry, Chinese Academy of Sciences, Beijing, China
- ⁵ University of Chinese Academy of Sciences, Beijing, China

based on the characteristics and feedback of students [4].

The knowledge tracking task can be expressed as a model of the states $S = \{s_1, s_2, \dots, s_{t-1}\}$ of knowledge of students by observing the sequence of past interactions $X = \{x_1, x_2, \dots, x_{t-1}\}$ of the students, and predicting the possibility of correct answers $p_t(y_t = 1|q_t, X)$ to the new exercise. Knowledge composition is a collective term for skills, concepts, problems, etc., which is used to describe knowledge. It is usually assumed that the learning process can be divided into two stages: (1) After each exercise, students learn and memorize related knowledge, and use a set of variables to summarize the levels of the hidden knowledge corresponding to the past exercise $x_t \sim f(h_t)$. (2) Students use the already learned knowledge to deal with new exercises, and predict the performances on the new exercises through the mastery of the hidden knowledge $h_t \sim g(h_{t-1})$. Input data is the sequence of past interactions, and each exercise is a tuple $x_t = (q_t, r_t)$, where q_t is the question at time t , and r_t is the correct answer to the question, usually binary.

However, it is inherently difficult to simulate the learning process of humans with simple numerical values [5]. With the explosion of online education and the increasing complexity of courses, complex knowledge components and rich semantic relationship between coursework and textbooks are encountered, which are collectively called as the complex background knowledge. This complexity poses two challenges to track knowledge accurately: (1) Sparse curriculum problem: It exacerbates the uneven distribution of interactions. For example, in order to avoid boring repetition, pedagogy experts design a variety of exercises, but all of them are not clicked by students. (2) Insufficient diagnosis problem: It produces many invalid diagnostic results. For example, since the mapping between knowledge and exercises is broad, many unimportant or difficult candidate exercises make learning suggestions difficult to implement. The question is how to capture the complex background knowledge for modeling the proficiency accurately?

Traditional probability models mainly include Bayesian Knowledge Tracing (BKT) [6, 7] and item response theory (IRT) [8]. BKT [7] uses binary state variables to describe knowledge mastery and use bayesian belief network to model knowledge updating. Its mechanism of state transition can be regarded as a simple noise binary channel, which is parameterized by the possibility of guessing, sliding, learning, and forgetting. On the other hand, IRT uses a value vector to describe knowledge mastery, model the interaction between the knowledge level and difficulty level through an item reflection function, and to predict the future performances of students. These early methods use strong assumptions for decomposing the background knowledge into a series of hierarchical knowledge components. Each

exercise uses one or more knowledge components, which is conducive to the derivation of the sparse knowledge as Fig. 1a. They rely on the concept of map predefined by pedagogical experts. Under the condition of high-quality expert experience, they can effectively estimate the levels of mastery of students' predefined knowledge, and explain the results of knowledge tracking and the evolution process.

The latest research focuses on how to model the learning behaviors in large-scale complex historical interactions through deep learning methods. A deep learning method named DKT [5] integrates the modeling of the long-term and short-term memory units (LSTM) [9] of knowledge evolution in the recurrent neural network (RNN). The knowledge state of a student is represented by one summarized hidden high-dimensional continuous vector, which encode the temporal behavior of information gathered in the past as Fig. 1b. Since the LSTM modeling unit is a nonlinear complex function, it is more flexible than affine transformation and can capture subtle differences in complex interactions. Although DKT substantially improves the predicting performance compared with the earlier probabilistic models, it is difficult to find specific patterns and track their evolutions. Dynamic Key-Value Memory Networks(DKVMN) [10] uses an external storage matrix, based on Memory-Augmented Neural Networks(MANN), to represent knowledge and extend DKT. The key matrix stores the correlations between exercises and concepts. The value matrix stores the correlations between the evolving exercises and the states of knowledge as Fig. 1c. The knowledge evolution is modeled by the "read and write" process. Since the memory matrix can increase capacity, update parts, and save relationships, it can model specific concepts gained from the recent exercises in long sequences. In general, the present KT research rarely discusses the important influence of complex background knowledge, and mainly focuses on the complex learning behaviors of humans. Deep learning methods have limited interpretability, and RNN-based methods, such as DKT and DKVMN, have poor generalization effects in dealing with sparse courses [11]. In short, the complex background knowledge poses new challenges to knowledge tracking. Traditionally, only human super teachers can handle such scenarios.

In response to the above challenges, we propose a new knowledge tracking method based on multi-hierarchical capsule graph neural network, called Caps-HAGKT, to capture the rich semantic relationships implicit in complex background knowledge. It transforms the traditional KT task into a time series node-level classification problem on a heterogeneous graph [12–14]. The new computational problem is based on three core assumptions: (1) Complex background knowledge can be regarded as a graph, called the knowledge structure, which is composed of a series of knowledge components, and is multi-hierarchical and

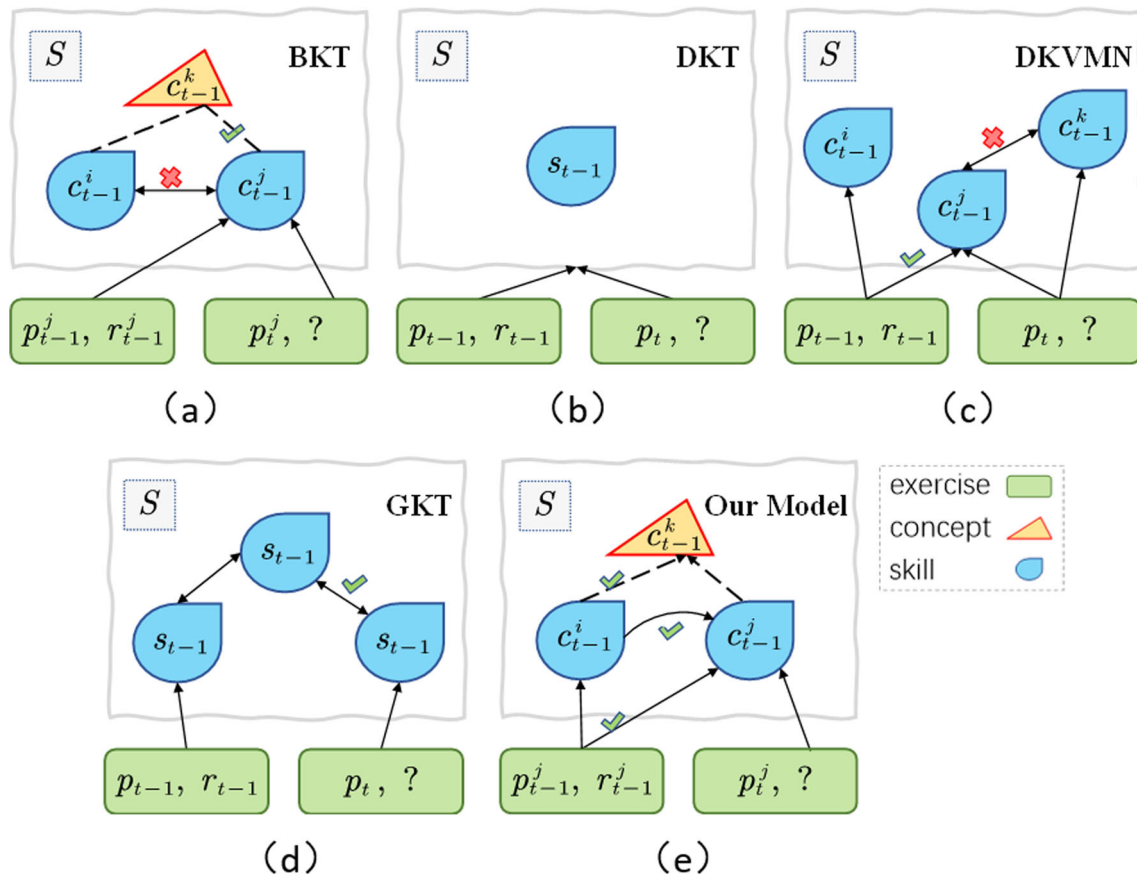


Fig. 1 Differences in the states of knowledge of students, as well as the relationships among BKT, DKT, DKVMN, GKT and our proposed model. Tuple (p_t^i, r_t^i) is exercise of learner. c_t^i and s_t^i are the concept states of low level and high level, respectively. The dependency

information marked by “sign in green color can be captured in the model, while that marked by “X” sign in red color cannot be captured. It is seen that our model can extract rich semantic relationships

multi-skilled [7], where each component corresponds to a concept of knowledge available for learning. There are rich knowledge semantic relationships between concepts, which are different types of shared dependent information, including prerequisites [15], attributions, implicit multiple skills [7], and long-term dependencies of different semantics [3, 10]. (2) The multi-hierarchical heterogeneous graph can represent the knowledge state of learners over time. The knowledge state describes the overall proficiency of all knowledge concepts, and the mastery of concepts is called the concept state. The low-level concept is a sub-part of the high-level concept, and the high-level concept state describes the overall mastery level of all states of the low-level concept. (3) Knowledge update is affected by the multi-hierarchical heterogeneous graph, and the state of the update concept also depends on other neighbor concepts. As Fig. 1e, for completed exercise q_t , the model actively discovers the low-level concepts i, j , and the high-level concept k latent in the complex background knowledge. Then, update the corresponding concept state of c_{t-1}^i, c_{t-1}^j , and

c_{t-1}^k . Additionally, in the case of the problem of sparse curriculum, Caps-HAGKT provides the relational inductive biases by using graph-structure data [16], which helps to enhance the ability of inductive speculation on future performance. In the case of the insufficient diagnosis problem, Caps-HAGKT provides interpretable knowledge tracing, which can trace sub-knowledge components or previous knowledge that affects the mastery of knowledge concepts. It is very important to provide learning suggestions and strategies for weak learners.

In Caps-HAGKT, we use multi-hierarchical knowledge capsule network, which is composed of the primary knowledge capsules, multi-hierarchical high-level capsules and routing mechanism. The primary knowledge capsules are generated by multi-head attention, which can model stable multi-skill relationships and knowledge features of different neighborhoods. A new arrangement of high-level knowledge capsule layers are designed, which can tracing all kinds of latent knowledge from abstract to concrete. The routing mechanism is used to extract the latent knowledge

structure (such as attribution relationship) between the high and low level knowledge capsules. It sends the detailed knowledge attributes to the appropriate high-level knowledge capsules. The multi-hierarchical knowledge capsule network can trace sub-knowledge components deeply. Note that it was difficult to trace latent knowledge concepts (such as GKT, DKT) in previous works. In addition, we use a graph network of knowledge capsules to capture the latent knowledge structure (such as the prior relationship) among the knowledge capsules at the same level, and an external memory matrix to store the corresponding knowledge state of each knowledge capsule. Static concept matrix stores fixed knowledge tags, while dynamic knowledge matrix updates knowledge that evolves over time. This kind of learning process is similar to human memory [17]. The ability to capture rich semantic information could not be provided in previous works (such as DKT and BKT). In Caps-HAGKT, we also use the semantic attention of knowledge capsules to integrate knowledge features with different semantic importance. A conceptual meta-path is used to capture the multi-hop compound semantic relationship between exercises and problem instances. The experimental results show that the proposed Caps-HAGKT can effectively utilize the latent knowledge structure information in the complex background knowledge, and model accurate knowledge state for the knowledge tracking of prediction and interpretability.

Finally, we summarize the main contributions of this research work as follows:

- (1) It provides a wealth of knowledge and semantic relationship information, and can improve the performance of knowledge tracking tasks, especially suitable for sparse practice interactive scenarios.
- (2) It improves the interpretability of knowledge tracking, and helps to provide targeted suggestions for learner's learning weaknesses. A novel multi-hierarchical knowledge capsule network is proposed for evaluating deep sub-knowledge components. The graph neural network of knowledge capsules is proposed to track the prerequisite knowledge and simulate the learning process based on memory for updating the knowledge state.
- (3) The multi-level heterogeneous graph structure latent in the complex background knowledge is modeled in an end-to-end manner. There is no need any pre-definition graph of experts is required. The model can automatically discover the semantic relationship of rich knowledge from the text and learning behavior. It also can learning graph structure itself with the KT main task in parallel.
- (4) Strengthen the ability to capture long-term dependent information in the sequence of historical interactive

data. Update the knowledge representation by adding the external memory matrix to improve the capacity of the model memory and the ability for expressing knowledge state. Moreover, the long historical experience, spread through the conceptual meta-path, helps to speculate on the recent performance.

2 Related work

2.1 Knowledge tracing

To tackle problems involving the knowledge of tracing task, researchers have developed various approaches from probabilistic models to deep learning-based models.

Probabilistic models BKT [7] is a highly constrained probabilistic model [3], which can trace the performances of specific concepts based on the concept map defined by domain experts. BKT assumes that knowledge can be perfectly decomposed into knowledge components, and uses binary values to indicate its mastery. Many of the following studies presented variations by personalizing learning [6], enriching information [18] and diversifying courses [19]. IRT [8] uses the characteristics of difficulty levels and the ability of students to predict the performances of specific concepts, which have strong interpretability. A logistic regression model is used as an item response function to evaluate changes in knowledge. Many improvements are made to gain greater expressiveness e.g. additive factor models [20], performance factor analysis [21], hand-crafted [22]. However, these methods are difficult to model complex human learning interaction behaviors by using only simple values, and they rely on probability maps predefined by experts.

Deep learning-based models DKT [5] is the first method to use deep learning, which breaks the strong assumption of perfect decomposition of knowledge and binary state, done specifically LSTM [9]. It uses a single hidden state to represent the summary of historical learning interaction. Experiments in [5] showed that DKT is substantially better than traditional probability-based methods. There are many extensions. Some research incorporates the prerequisite relations between pedagogical concepts [15]. Some researchers focuses on explainable and personalized performance [23]. Using the Key-value external memory matrix, DKVMN [10] expanded DKT's interpretive power of knowledge. It uses separate "read" and "write" processes on this external matrix to the evolution of specific concepts. DKVMN is a special kind of RNN. It can increase storage capacity by using the external memory matrix and encourage for transitions of local states. [24–26] makes

inference faster and enhances the capacity of capturing long-term dependencies in an exercise sequence by using a modified LSTM with hops. However, these RNN-based methods do not consider the rich semantic relationship between implicit concepts and provide high-dimensional implicit variables that are difficult to understand. In order to overcome these difficulties, we construct the learning material as a knowledge graph, and transform the sequence prediction problem into a time-series node-level classification problem.

SAKT [27] is the first knowledge tracking method using the attention mechanism. It models the entire exercise interaction sequence and uses the knowledge features of a few related exercises. AKT [28, 29] extends SAKT, which replaces fixed location information with forgetting coefficients, and considers the contextual knowledge characteristics of similar problems. RKT [30, 31] also uses exponential decay coefficients and integrates contextual information. However, these attention-based methods directly use a few of past related exercises to predict student performance, which is difficult to model the knowledge level of specific concepts. Through the capsule network, we can not only model specific concepts but also model all kinds of knowledge components.

GKT [14] finds that there is a latent graph structure in the coursework. It uses Graph neural networks (GNN) [32] to calculate relational induction bias to track the evolution of knowledge. Exercises are mapped to knowledge concepts, as nodes, and the shared relations between concepts are taken as edges, which affect the updating of knowledge status. GKT can extract readable knowledge structure and use knowledge semantic relationship information, which is very interesting and worthy to our reference. However, the existing work of GNN does not consider the complexity of the knowledge structure. We model a hierarchical knowledge structure and integrate rich semantic relationship information.

2.2 GNN and meta-path

Recently, graph neural networks (GNN) [32] has received widespread attention due to its powerful ability to process and express irregular data. Different generalization frameworks and important operations of GNN have been developed with successful results in various research areas, such as social science [33] and natural science [34], natural language processing [35], computer vision [36] and recommendation systems [37]. Similar to BKT, GKT represent knowledge as hierarchical knowledge components, which help to obtain the interpretability of the predicted results. Therefore, complex background knowledge can be regarded

as a special graph structure, which is widely known as heterogeneous information network [13] with multiple types of nodes and edges.

Meta-path [38] is a structure widely used to capture semantic information and to describe a composite relation connecting two entities. Wang et al. [12] presumes that heterogeneous graphs involve complex semantic information, and different Meta-paths can reflect different semantic relationships. Closely related to our research, historical experience is very important for predicting future performance of a student as the captured important semantic relationship information can improve the accuracy of a knowledge model.

2.3 CapsNN

CapsNN [39] is a neural network used to recognize and classify complex entities, such as pose and deformation. A capsule is a set of neurons used to represent different characteristics, which captures the hierarchical posture relationship between various parts of an entity during the dynamic routing of the capsule. Many latter studies mainly focused on reducing the computational scale, and expected to replace CNN, and improve capsule quality [40] and routing strategies [41]. [42] extended CapsNN to graph data, where node capsules were used to fuse rich detailed features so as to generate sub-graph representations. This provides us with a lot of inspiration for tracking high-level concepts in complex background knowledge.

3 Problem definition

The knowledge tracking task can be understood as the modeling of knowledge of students and its evolution based on historical interactive exercises. It can also predict the possibility of answering new questions correctly. We further assume that the exercise question Q involves C implicit concepts, and the concept hierarchy is divided into a total of L types. A concept determines the type of relationship, which includes the prerequisite relationship between concepts, attribution relationship between concepts at different levels, and multi-skill relationship between problems and implicit concepts. The complex background knowledge can be regarded as a multi-level heterogeneous graph $G(A, V)$, where node V represents the implicit knowledge concept C and edge A represents the dependency relationship in concept sharing. In contrast, the knowledge state of students is the sum of concept states, which describes the overall mastery of all the concepts. We use the static concept matrix M_k for storing the knowledge label,

and the dynamic knowledge matrix M_v is used to update the independent concept state.

Embedding layer: We encode questions and exercise interactions, and integrate rich semantic representations by concatenating text vectors. Each interaction is represented as a tuple $x_t = (q_t, r_t)$, there are Q unique question tags and knowledge characteristics of a D dimensional exercise. Therefore, the question embedding matrix E is used to obtain the indices representing questions. There are Q question embedding vectors. Similarly, we train the exercise embedding matrix M , M_{s_i} for each element, s_i in the exercise sequence. There are $2Q$ question-response embedding vectors.

Text of material encoding: [43, 44] finds that a simplified use of one-hot to express exercises by ignoring texts causes huge information loss. A large amount of text contents in the learning and teaching materials describes contextual semantic relations and teaching logic. So we sort out each question and concept tag into a sequence of text contents composed of words, and transformed into corresponding text vectors with Bert [45]. Here, we used Bert-as-service¹ open source project to build Bert, and loaded the uncased Bert-base model². This type of Bert contains 12 layers of networks and 768 hidden variables.

Location encoding: The location information is very important for capturing a reasonable evolution of knowledge. The state of knowledge at a particular time instant should not show any wavy transition. We use the one-hot coding to locate the order of the learning interaction sequence. We define the position embedding matrix P to store the locations and time related information, where a row represents the positioning sequence of the exercise. We then add the row i of the position embedding matrix P_i to the exercise embedding matrix for generating past exercise embedding $\hat{M}$.

4 Proposed Caps-HAGKT method

We propose a new method, namely the multi-heterogeneous capsule graph network (Caps-HAGKT), which can tracking a variety of implicit knowledge composition and their complex relationships without any expert pre-definition. Figure 2 shows the entire structure of Caps-HAGKT. It consists of three blocks:

- 1) Multi-hierarchical knowledge capsule network: A multi-head self-attention is used to generate primary capsules, and a capsule routing is used to discover the latent relationship between knowledge levels.
- 2) Graph network of knowledge capsules with external memory matrix: It uses GNN to extract latent shared dependencies between the knowledge components at the same level. It also updates the knowledge state capsule stored in the external memory matrix.
- 3) Semantic attention of knowledge capsules: A meta-path is used to capture a variety of knowledge semantics, and also to integrate important knowledge representations for prediction.

4.1 Multi-hierarchical knowledge capsule network

In order to tracing the deep key component parts of the knowledge. Instead of a hierarchical structure predefined by experts, we use a network of multi-hierarchical knowledge capsules to capture the information about the latent knowledge structure between levels (such as attribution dependency). It is composed of primary knowledge capsules, a series of high-level knowledge capsules, and capsule routing.

Firstly, we designed multiple layers of high-level knowledge capsules, where each layer corresponds to a particular kind of knowledge component. The knowledge is decomposed and refined layer by layer in the high-level knowledge capsule. It conforms to the key concepts of the positioning process from the abstract to the concrete. As shown in Fig. 3, there are 3 kinds of knowledge components and 3 layers of high-level knowledge capsules, and knowledge mastery can be tracked along the knowledge logic from the “four arithmetic operations” to the “decimal addition”. This is similar to the gradual discovery of shortcomings in knowledge by human education experts. The closer to the primary capsule layer, the wider the knowledge domain and the smaller the scale of knowledge components, and vice versa. Therefore, this kind of knowledge tracking results with clear knowledge logic can be obtained.

Secondly, the primary knowledge capsules is generated by the multi-head attention, which can obtain a stable multi-skill relationship between problems and skills in the exercise sequences. The difference is that we extract multi-aspects knowledge properties from different heads and save them in the form of capsules. This means that the knowledge capsule itself has a clear physical meaning and is interpretable, that is, the latent knowledge feature of a certain aspect of the historical exercise interaction. The calculation process is as follows.

¹Bert-as-service is retrieved from <https://github.com/hanxiao/bert-as-service>.

²The uncased Bert-base model is retrieved from <https://github.com/google-research/bert>.

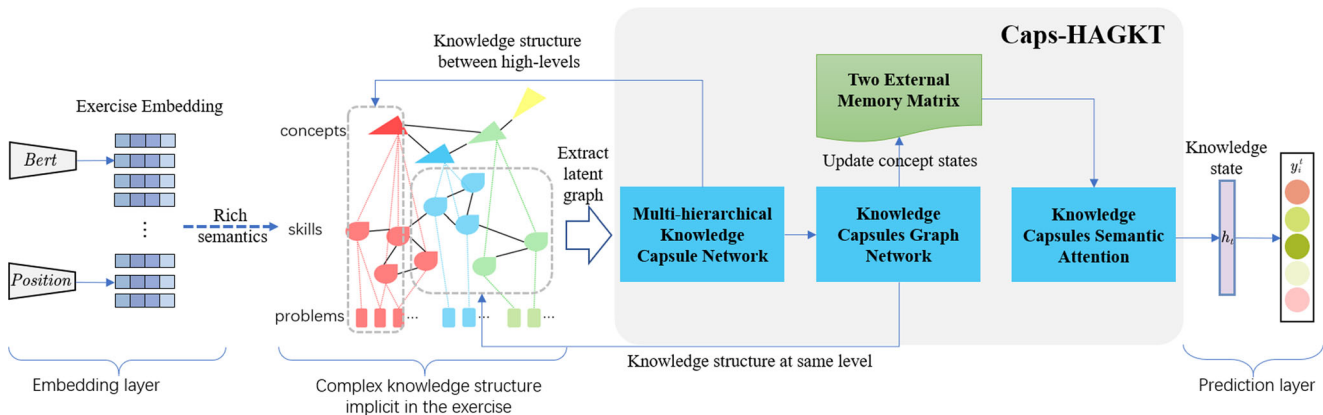


Fig. 2 Framework of Caps-HAGKT. It can automatically extract complex knowledge structures without any expert pre-definition

(1) The past exercise embedding $\hat{M}$ of the participating students and the unique question tag $\hat{E}$ are linearly mapped into different vector spaces of query, key, and value through the projection matrix. The weight coefficient of the multi-skill relationship is further calculated by taking the softmax function of the scaled dot product between the key vector $K(j)$ and each fixed memory vector $M^{k1}(j)$. Its specific definition is as follows:

$$\alpha_{ij}^t = \text{soft} \max_{N_i} \left(\frac{M^{k1}(i) K_i^T(j)}{\sqrt{d}} \right) = \frac{\exp \left(\frac{M^{k1}(i) K_i^T(j)}{\sqrt{d}} \right)}{\sum_{k \in N_i} \exp \left(\frac{M^{k1}(i) K_i^T(k)}{\sqrt{d}} \right)} \quad (1)$$

Where, α is the weight coefficient of the multi-skill relationship, which represents the correlation between the practice label feature and the primary skill index M^{k1} . We inject graph structure information through masked attention

weights, and calculate only exercise $j \in N_i$ in the skill neighborhood, where N_i is some neighbors of exercise node of skill i in the graph structure.

(2) After obtaining the normalized multi-skill relationship weight coefficient, the primary skill node represents the linear combination of historical practice knowledge features in the neighborhood, which is calculated by the graph-based aggregation operation. The aggregated features u_i^t are activated through the following nonlinear function:

$$u_i^t = \sigma \left(\sum_{k \in N_i} \alpha_{ij}^t W_0 V_i^T(j) \right) \quad (2)$$

Since the multi-head attention can independently observe different neighborhood features, it not only makes the multi-skill relationship more stable, but also inspires us to obtain rich and detailed features of sub-component skills from different receptive domains for generating high-level concept capsules. As in [46], using K independent attentions for performing the above transformation, the

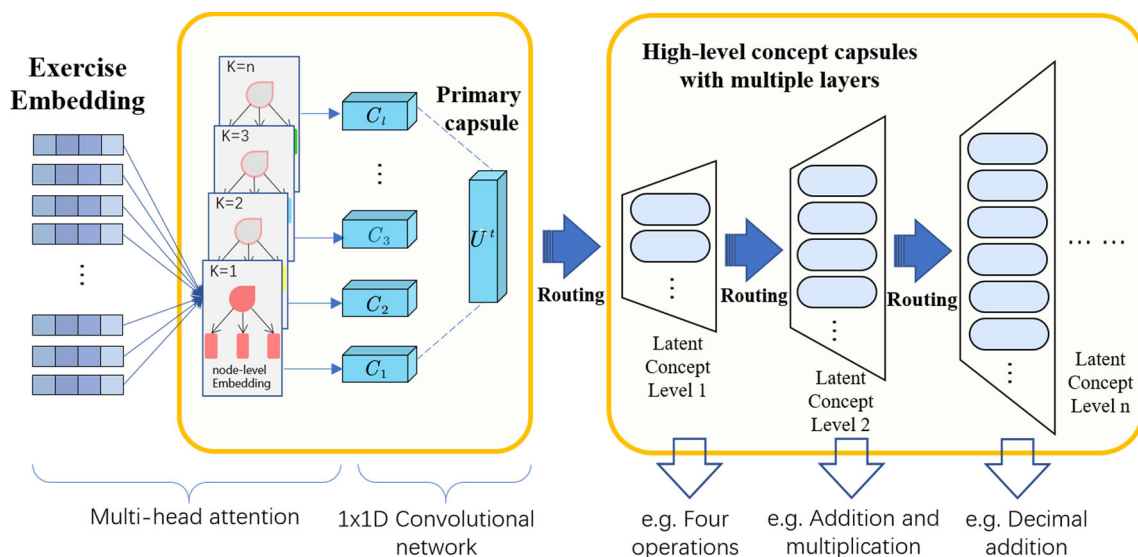


Fig. 3 Explanation of multi-hierarchical Knowledge Capsules

feature representation of the multi-head is calculated as follows:

$$u_i^t = ||_{k=1}^K \sigma \left(\sum_{j \in N_i} \alpha_{ij}^k W_0^k V_i^T(j) \right) \quad (3)$$

Where, K is the number of attention heads. T and $||$ denote the transpose and connection operations, respectively. The non-linear activation function σ is obtained by using LeakyReLU. α_{ij}^k is the weight coefficient of the multi-skill relationship calculated by the k -th attention mechanism. W_0^k is the corresponding trainable linear transformation weight matrix.

(3) In order to preserve the detail of the rich attributes of different neighborhoods of sub-component skills, we express the extracted features in a capsule form to generate high-level concept capsules. Specifically, N groups of primary skill node capsules are generated, where each group is expressed as $U_n^t = \{u_{11}^t, \dots, u_{1C1}^t, \dots, u_{NCL}^t\}$. s_{nc} represents a characteristic instance of a primary skill attribute. C_l represents the number of knowledge attribute features in the l -th skill neighborhood. Note that, in this setting, the dimension of the capsule vector is K , which means that the number of attention heads is the same as the size of the capsule.

Finally, we use the capsule routing to calculate the weights of the knowledge capsules between layers, so as to propagate the captured knowledge features of a certain aspect to suitable high-level knowledge capsules in the next layer. We use these coupling coefficients r_{ij}^L in depicting the dependency between the high and low levels of knowledge capsules, such as the relative attribution relationship. The capsule routing that we adopted includes the following 3 key steps:

$$v_{ij}^L = W_{ij}^L \cdot u_i^L \quad (4)$$

$$r_{ij}^L = \text{Softmax} \left(s_j^{L+1 T} \cdot v_{ij}^L \right) \quad (5)$$

$$s_i^{L+1} = \text{LayerNorm} \left(\sum_i r_{ij}^L v_{ij}^L \right) \quad (6)$$

Where, u_i^L is the low-level knowledge capsule vector; v_{ij}^L is the feature representation that may be related to the high-level knowledge capsule and they are extracted from the low-level knowledge capsule; The coupling coefficient r_{ij}^L is updated through the correlation coefficient between the high-level knowledge capsule vector s_i^{L+1} and the low-level knowledge capsule intermediate vector v_{ij}^L ; The high-level knowledge capsule vector s_i^{L+1} is updated by the coupling weight coefficient r_{ij}^L and the low-level knowledge capsule intermediate vector v_{ij}^L ; Capsule routing can be achieved by

following [39] and [47], we recommend the latter one which has smaller parameter scale.

4.2 Knowledge capsules graph network with external memory matrix

The relationship between high and low level knowledge capsules is a whole and part, and routing is mutually exclusive in the parent knowledge capsule. However, the concepts of knowledge are usually not isolated, and the recommended effective teaching requires the understanding of the mastery of relevant knowledge, especially some prerequisite concepts. Therefore, in order to trace other key components of knowledge which may affect the concepts of learning, we use the graph network of knowledge capsules with an external memory matrix to model the implicit knowledge structure between the knowledge capsules at the same level and to update the knowledge state capsule. The concept states update process is as shown in Fig. 4.

Firstly, we use a shared attention to calculate the weight coefficient of the normalized relationship between concepts as follows:

$$\beta_{ij} = \text{softmax} \left(\frac{\exp \left(\text{LeakyReLU} \left(\tilde{a}^T \cdot [W_2 M_i^k || W_2 M_j^k] \right) \right)}{\sum_{x \in N_i} \exp \left(\text{LeakyReLU} \left(\tilde{a}^T \cdot [W_2 M_i^k || W_2 M_x^k] \right) \right)} \right) \quad (7)$$

Where, W_2 is the parameterized shared weight matrix for linear transformation. The weight coefficient of the shared attention can be calculated by the dot product operation between the trainable weight vector $\tilde{a}^T$ and the splicing vector $[|| \cdot]$. The activation function is LeakyRelu, and its input slope is -0.2. The normalization function uses softmax to ensure that the weight coefficients are comparable. The static concept matrix M^k saves semantic relationship information, and its each row represents a concept index. Only the neighborhood knowledge $j \in N_i$ of concept i is calculated.

After obtaining shared relationship weight coefficients, We update the knowledge state capsules of learners over time. At each time stamp, the knowledge matrix M_i^{vln} is updated by using the following formula:

$$M_i^{vln}(i) = \begin{cases} \text{squash} \left(\frac{1}{K} \sum_{k=1}^K \sum_{j \in N_i} \beta_{ij}^k W_2^k s_j^t \right) & (l_n \neq l_1) \\ \text{squash} (W_{l_1} \cdot u_i^t + b_{l_1}) & (l_n = l_1) \end{cases} \quad (8)$$

Specifically, in order to minimize the attention variance, we use K attention head mean aggregation operations to calculate the high-level conceptual knowledge embedding, and update the knowledge memory matrix accordingly. We use squash [39] to activate the concept capsule.

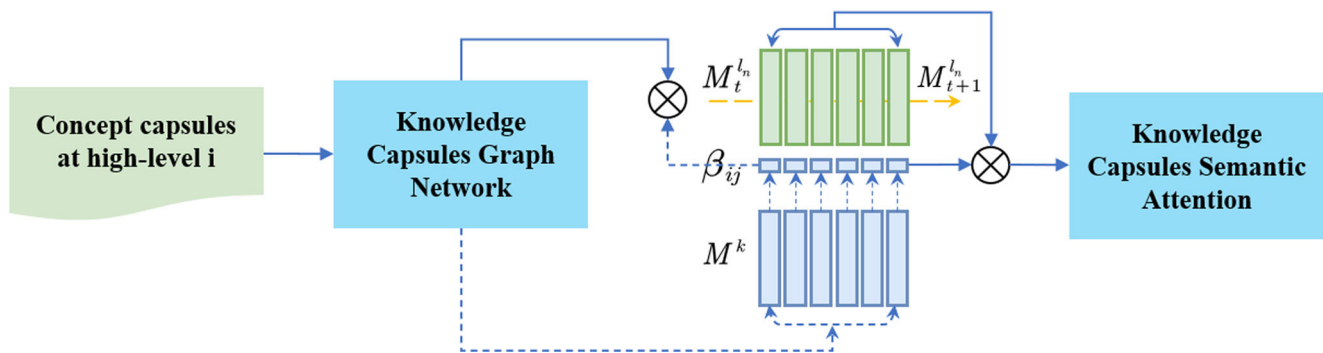


Fig. 4 Explanation of the concept states update process by knowledge capsule graph network

Knowledge state of explicit graph structure: Our model can use a graph to depict the knowledge state of a learner. The mastery of each concept and the relationship between the concepts will help to adopt the targeted learning strategies according to the strengths and weaknesses of the learners knowledge. The knowledge state of a learner is composed of the capsules of the concept state and knowledge structure. We convert the capsule vector into an understandable probability value by calculating the length of the normalized vector modulus as follow:

$$P(feature) = ||squash(s_j)|| \quad (9)$$

Where squash [39] is used to ensure that capsule vectors are comparable.

4.3 Knowledge capsules semantic attention

Different hierarchy of knowledge capsules have different knowledge semantics. In order to integrate rich semantic information, We use conceptual meta-paths to capture the

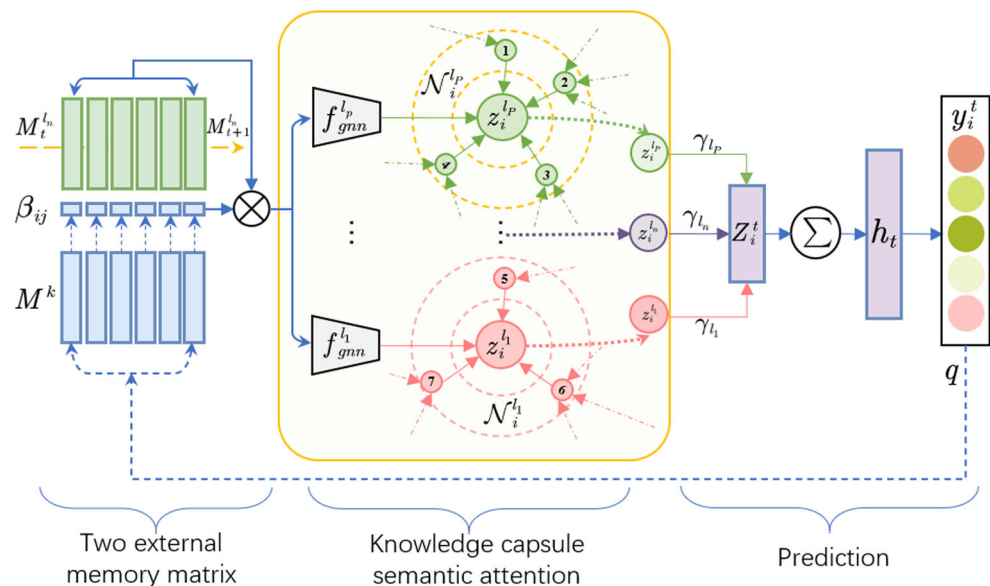
knowledge-semantic compound relationship and use attention automatically learn the importance of different conceptual meta-paths. We call this mechanism as knowledge capsules semantic attention, as show in Fig. 5.

(1) Each category level of knowledge concept corresponds to a concept meta-path and provides a semantic relationship. There is a total of P types of levels, which can be expressed as $L = \{0, l_1, l_2, \dots, l_P\}$. Therefore, the neighborhood features on the conceptual meta-paths l_n are aggregated to obtain the corresponding problem instance capsule as follows:

$$z_i^{l_n} = f_{gnn} \left(A_{N_i}^{l_n} s_j W_3 \right) \quad (10)$$

Where, $z_i^{l_n}$ represents the embedding knowledge state of problem instance i on the l -th conceptual meta-path. $A_{N_i}^{l_n}$ is the adjacency matrix of the l -th conceptual meta-path, which indicates that node j is on the conceptual meta-path and belongs to the neighborhood N_i of node i . s_j is the knowledge concept capsule vector, which represents the knowledge state of concept j . W_3 is a projection matrix

Fig. 5 The process of integrating different semantics by knowledge capsule semantic attention



and a learning parameter that can be optimized during the training. The graph neural network f_{gnn} is used to aggregate the neighborhood features of the concept element path, and $l_n = 0$ represents the characteristics of the problem instance.

For a series of conceptual meta-paths $\{l_1, l_2, \dots, l_P\}$ of the multi-hierarchical heterogeneous knowledge state graph, we obtain the knowledge embedding of the problem instance through the knowledge memory matrix $M^{v_{l_n}}$ loading the recovery process, which is expressed as a group of capsules $\{Z_{l_1}, Z_{l_2}, \dots, Z_{l_P}\}$.

(2) After obtaining the problem instance capsules of different conceptual meta-paths, we further determine the importance of each conceptual meta-path to the problem instance by calculating the correlation between these different semantic vectors and the problem instance query vector q_t . The weight coefficient of the conceptual meta-path semantic relation is normalized as follows:

$$\gamma_i = \text{soft max}_P \left(\frac{1}{|\mathcal{N}_i|} \sum_{i \in \mathcal{N}_i} q^T \cdot z_i^{l_n} \right) \quad (11)$$

(3) After learning the weight coefficients, we fuse different semantic relationship information to restore the final problem instance capsule as follows:

$$Z_t = \sum_{i=1}^P \gamma_i \cdot Z_{l_i}^t \quad (12)$$

(4) We splice the problem instance capsule with the input exercise key vector, and feed the result to the prediction layer. The summary vector h_k^t integrates the prior difficulty of the problem instance and the corresponding knowledge mastery of students.

$$h_t = \text{concat}(z_t, k_t) = [z_t, k_t] \quad (13)$$

4.4 Prediction

The model finally gives the probability of students answering each question instance in the next learning activity:

$$y_k^t = \text{squash}(W_4 h_k^t + b_k) \quad (14)$$

Where, W_4 is a parameterized weight matrix which is common to all problem instance nodes, and b_k is the bias term of the problem instance node k . It is worth mentioning that the final predicted output (the capsule vector) y_k^t is not a scalar. We use the length of the capsule vector to indicate the probability of correctness of each problem instance. Therefore, the non-linear squashing function is used to activate the capsule vector, instead of using the commonly used the sigmoid function.

4.5 Optimizations

The training optimization goal of the model is to minimize the negative log likelihood between the observed performance of real students and the model prediction result. During the training process, the initial value of the memory matrix M^k , $M_t^{v_{l_n}}$ and other learning parameters are jointly learned by minimizing the standard cross-entropy between the predicted result capsule vector modulus $\|p_t\|$ and the real label r_t .

$$\mathcal{L} = - \sum_t (r_t \log \|p_t\| + (1 - r_t) \log (1 - \|p_t\|)) \quad (15)$$

In addition, the default hyper-parameter of the major network structure of Caps-HAGKT is shown in Table 2, including the graph network layer, capsule layer and semantic layer. In order to trace the interpretable knowledge, the model needs to be set up according to the actual requirements and datasets. For example, the assist2009 dataset has 2 knowledge levels, which use 10 and 110 high-level capsules, respectively.

5 Experimental setup

5.1 Datasets

We test the accuracy of the proposed model for knowledge tracking tasks on different types of datasets, including 1 simulation dataset, 4 public real datasets, and 1 private business dataset. The complete statistical information of all the data sets is presented in Table 1.

Synthetic-5³ consists of two parts: training set and test set. It simulates 4000 virtual students and a total of 200,000 historical learning interaction records (i.e. tuple(q_t, r_t)) on 50 problem instances. These problems stem from a total of 5 types of implicit concepts with different degrees of difficulty. As the ground truth, it is very helpful to visually evaluate the effect of the concept discovery.

ASSISTments2009⁴ is a public dataset provided by a real online teaching aid platform. It is widely used in knowledge tracking tasks. Many studies used it to evaluate model performance. A total of 325,637 historical learning interaction records were collected from 4,151 real students for 110 question examples. The sparse problem of this data set is more serious, and its data density is 0.06.

³The Synthetic datasets are retrieved from <https://github.com/chrispiech/DeepKnowledgeTracing/tree/master/data/synthetic>.

⁴The ASSISTments2009 datasets are retrieved from <https://sites.google.com/site/assistmentsdata/home/assistent-2009-2010-data/skill-builder-data-2009-2010>.

Table 1 Dataset statistics

Dataset	#Students	#Problems tags	#Interactions	#Density
Synthetic-5	4,000	50	200K	1
ASSIST2009	4,417	110	325K	0.06
ASSIST2015	19,840	100	683K	0.05
ASSIST2017	686	102	942K	0.81
Statics2011	333	1,223	189K	0.31

¹ The density of each dataset is calculated by $\#Unique\ Interactions / (\#Users \times \#Problems\ tags)$

ASSISTments2015⁵ collected historical learning behaviors of more students, including real-world results of 19,840 students answering 100 question examples, and a total of 683,801 valid historical interaction records. In the preprocessing, it removes the records containing invalid responses. The density of the data set is 0.05 and the degree of its sparseness is very high.

ASSISTment2017⁶ is the most abundant data set, which was used in the Challenge of KDD-Cup. It contains a total of 942,816 historical interaction records, which describe the responses of 686 students to 102 question instances. The problem instances are densely distributed, with a density as high as 0.81.

STATICS2011⁷ is derived from a college-level engineering statics course, which is also a publicly available data set. It contains a total of 189,927 valid historical interaction records, involving answers of 333 real-world students to 1,223 questions. According to statistics, the data density is 0.31, which is relatively sparse.

All datasets are preprocessed as follows:

- (1) Clean invalid data, such as discarding meaningless historical learning behaviors with a length less than 2.
- (2) All data is divided into 3 parts, 70% of the historical interaction records are used for training, and 20% of the data is used as a test for selecting the best model architecture by optimizing hyper-parameters and performing early suppression. The remaining 10% data is used to verify the effectiveness of the model.
- (3) Concept tags are transformed into word sequences, and these text contents are encoded into a 768-dimensional vector with Bert. Generally, concepts with attribution or prerequisite relations have high semantic similarity.

⁵The ASSISTments2015 datasets are retrieved from <https://sites.google.com/site/assistmentsdata/home/2015-assistments-skill-builder-data>.

⁶The Synthetic ASSISTment2017 are retrieved from <https://sites.google.com/view/assistmentsdatamining>.

⁷The Synthetic datasets are retrieved from <https://pslcdatashop.web.cmu.edu/DatasetInfo?datasetId=507>.

5.2 Implementation details

5.2.1 Metrics

We compare the proposed Caps-HAGKT with a series of baseline works as introduced in the status analysis, which are BKT, DKT, DKVMN, and GKT.

It is considered that the knowledge tracking and prediction task have binary classification. For example, students get 1 point for a correct answer and 0 point for an incorrect answer, indicating positive and negative samples, respectively. Accordingly, the area under the ROC curve (AUC) is used to quantitatively measure the performance of the model. The larger the value of AUC, the higher is the predicting accuracy of the model. If the value is 0.5, the model performance is equivalent to random guessing or coin tossing only.

In order to ensure the effectiveness of the performance indicators of the reported model, 5 training and evaluation processes were carried out. Using 5 different initialization parameters, each initialized parameter is allowed to perform 500 epochs training process. The model having best performance on the test set is saved, and the average value of the test set evaluating index is returned.

5.2.2 Parameter setting

During the training process, the external memory matrix that saves the knowledge concept and state is constantly optimized. The concept matrix remains fixed during the test. The initial value of the knowledge matrix can describe the difficulty level of each concept. The specific implementation of the SGD algorithm is the Adam [48] optimizer with a momentum of 0.9 and a learning rate $\gamma = 0.01$. The threshold for stopping the model training function early is 100, that is, the training is stopped if the verification loss is not reduced in 100 epochs.

For the proposed Caps-HAGKT model, the PyTorch development platform is used to realize the entire model structure during the implementation period. A variety of scenarios $d = \{64, 128\}$ were tested for the batch size, while

for other baseline methods the hyperparameter settings were taken from the papers where those were proposed. The maximum sequence length is roughly proportional to the average motion label of each student. ASSIST2017 and STATICS use a fixed length of 500, ASSIST2009 uses a fixed length of 100, and Synthetic and ASSIST2015 use a fixed length of 50. If the length of the actual input sequence exceeds the set threshold, we decompose the entire sequence into multiple short sequences. If the actual input sequence is not enough to set the threshold, we use a pad to fill every short sequence whose length is less than a given fixed size.

In addition, the default hyperparameter of Caps-HAGKT's major network structure are shown in Table 2, including graph network layer, capsule layer and semantic layer. In order to achieve interpretable knowledge tracking, the model needs to be set up according to actual requirements and datasets. For example, the assist2009 dataset has 2 knowledge levels and uses 10 and 110 high-level capsules respectively.

6 Results and discussion

6.1 Performance prediction

Table 3 shows the performance comparison of Caps-HAGKT with the current state-of-the-art methods. Specifically, compared with the sub-optimal DKVMN results in STATICS, our model is found having an improvement of

2.56%. For ASSIST-2009, our model is well adapted to low-density data scenarios, where it is 3.69% higher than DKT on an average. Although ASSIST-2015 is not strongly sparse and short per capita behavior, Caps-HAGKT still achieves here an improvement of up to 2.40%. The AUC of our proposed model in ASSIST-2017 is also 2.6% higher than that of DKVMN. It is worth noting that although ASSIST-2017 has a large data scale and a high distribution density, all models do not perform well on it. This may be because of a weaker relationship between problem instances or a large difference in student behaviors. On Synthetic, each unique problem instance is derived from the same knowledge concept. It makes those models to perform well, which are good at capturing the relationships between problems, such as GKT and DKVMN. From the AUC results of all datasets, the new model has depicted a consistent improvement in the accuracy of tracking knowledge.

In short, the above experimental results prove that Caps-HAGKT can model complex student knowledge more accurately. In addition to that, we observed the following also: (1) Since deep learning can model complex interactive behavior patterns, it is better than Bayesian methods and it does not need to be oversimplified in knowledge composition and relationships. (2) Using both external memory matrix and GKT based on graph structure, DKVMN can capture latent concepts and relationships, which are more interpretable than in DKT. (3) In comparison to these works, our model can capture the rich semantic relations of the knowledge space and strengthen

Table 2 Comparison of prediction performance

Layer	Operation	Output Size
primary capsules	input dim=49x64x4, output dim=49x64x10 1x1 conv, stride=1, padding=0, Layer Norm	49x16x4x10
high-level capsules	Capsule Routing to 10 16x5-dim capsules 3x3 conv, stride=3, Layer Norm	16x5x4x10
GNN capsules	input dim=320, output dim=320 heads=2, share-attn, LeakyReLU	16x5x4x10
high-level capsules	Capsule Routing to 10 14x3-dim capsules 3x3 conv, stride=1, Layer Norm	14x3x4x10
GNN capsules	input dim=168, output dim=168 heads=2, share-attn LeakyReLU	14x3x4x10
class capsules	Capsule Routing to 10 4-dim capsules Layer Norm	10x4
meta-path capsules	input dim=64, output dim=64 meta-path=3, soft-attn, Layer Norm	10x4

¹ soft-attn is the soft attention mechanism. conv is the convolutional neural network. share-attn is the shared attentional mechanism

² This table show the default Caps-HAGKT

Table 3 Comparison of prediction performance

Dataset	BKT	DKT	DKVMN	GKT	Caps-HAGKT	Gain%
Synthetic-5	0.620	0.805	0.825	0.812	0.852	2.67
ASSIST2009	0.639	0.763	0.787	0.722	0.824	3.70
ASSIST2015	0.638	0.728	0.727	0.741	0.765	2.40
ASSIST2017	—	0.730	0.716	0.736	0.742	0.60
Statics2011	0.730	0.817	0.822	0.820	0.848	2.56
Average	0.653	0.769	0.776	0.766	0.806	3.05

¹ The reported performance are the best hyperparameter selection in each dataset and bold marks indicate the best results

the accuracy of tracking the knowledge level of students, especially to alleviate the problem of sparse courses. Figure 6 illustrates the ROC curves of these five models for each dataset.

6.2 Interpretability of prediction

People want to use a knowledge tracking technology to find out how students acquire knowledge in the learning process, and to have a deep understanding of the process of knowledge evolution. Intuitive description of changes in the knowledge status of students will help the students to adjust their learning goals themselves, help education experts

improve course design, and help the platform to provide personalized intervening suggestions. We visualized the knowledge state of students predicted by the model and evaluated its ability to explain the evolution of knowledge. This is a work of great significance.

6.2.1 Visualization of knowledge mastery

Since the interaction mode and conceptual relationship are difficult to analyze, DKT shows that a high accuracy of the model does not mean that the update of the knowledge status of students is reasonable and consistent. Therefore, we analyze and evaluate whether the model is

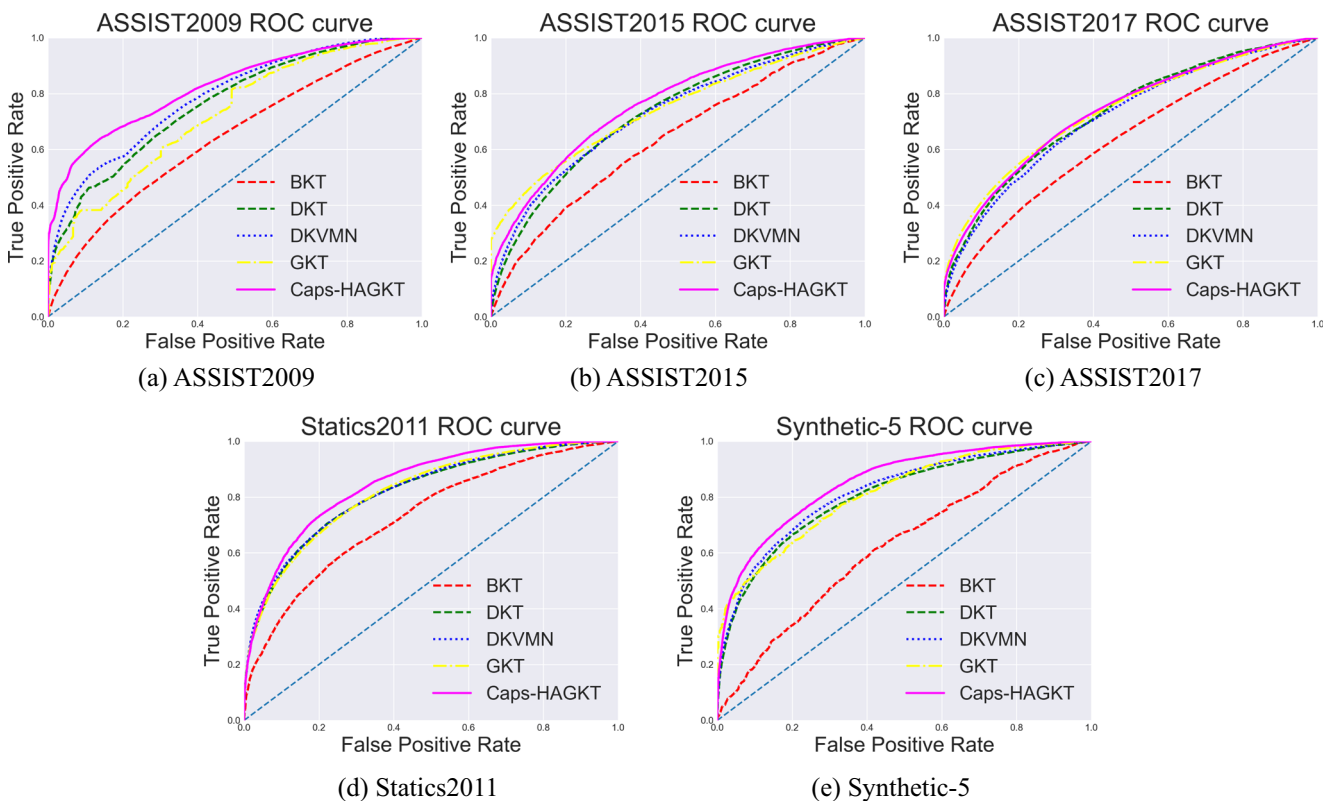


Fig. 6 The ROC curve results of the five models over five datasets: (a) ASSIST2009, (b) ASSIST2015, (c) ASSIST2017, (d) Statics2011, (e) Synthetic-5

interpretable from the following two aspects: (1) The update of the knowledge status of students should not violate common sense. (2) Knowledge evolution should not ignore dependencies.

We randomly selected the experimental output of the model on synthetic-5, and used Heatmap to visualize the evolutions of students' knowledge. As figure shows, during 50 consecutive learning interactions of one student, the model tracks the states of five unique skills behind the exercise sequence. As Fig. 7, the evolution process of knowledge shown by the new model is smooth and free of oscillation. For example, in the continuous wrong answers (31 to 34) and the continuous right answers (3 to 7), the change in the knowledge level of the student is continuous and reasonable. Moreover, it also shows the ability to update relevant knowledge. For example, when answering question number 24 correctly, the mastery of relevant skills for question number 51 and 54 is improved. These cases show that the results predicted by our model are more interpretable, indicating that it can better track the learning process of students.

6.2.2 Discover latent knowledge graph

In order to intuitively understand which knowledge components are the strengths and weaknesses of the learners, we visualized the knowledge structure of the students and evaluated the interpretability of the predicted results through case studies. Same as the previous work [10, 14], we only selected the Synthetic-5 and the ASSIST2009 as the benchmark datasets as the former simulates exercise records of the learner on 5 mutually independent latent concepts, and the latter has sufficient texts and a large number of real records of students. These texts and known classifications can be used as the ground truth and help to directly evaluate the ability of the model to interpret the predicted results. It is difficult to obtain the same in other datasets.

On the Synthetic-5 dataset, we visualized the clustering results through t-sne, as Fig. 8a. Obviously, courses are

divided into 5 categories, which means that the proposed Caps-HAGKT can tracing the 5 latent concepts behind the exercises and learning materials. Figure 8b is the corresponding knowledge structure, which shows that the dependency relationship among the 5 concepts is equilateral five-angle type, and the influence of any one concept is balanced. This is consistent with the setting of the dataset, which requires 5 concepts to be independent of each other. In addition, Fig. 8c is the overall knowledge structure learned by Caps-HAGKT, which shows two-hierarchical concepts. Specifically, the extracted multi-skill relationships and prerequisite relationships can be observed. The interesting thing is that the model can also find two courses, 4 and 33, which are far apart in time, but both belong to the implicit concept 54. However, these rich knowledge structures and the evolution of knowledge affected by graph are difficult to reveal by previous methods.

On the larger ASSIST2009 real dataset, we also visualized the clustering results and the corresponding implicit knowledge structure, as shown in Fig. 9a and b, respectively. Fig. 9c shows the overall knowledge structure, which describes three-hierarchical knowledge and rich semantic relationships between concepts. Additionally, in Fig. 9d, local connections and corresponding text labels are also shown. These reveal reasonable results, for example, related exercises 21,23,48 and 74 belong to the same implicit concept of Decimals. In short, the proposed Caps-HAGKT can extract implicit knowledge structures, use rich semantic relations, and provide interpretable knowledge tracing. This helps to provide recommendations and diagnoses for learners' knowledge weaknesses.

6.2.3 Ablation Study

We carry out a series of ablation studies, and further analyze how the proposed Caps-HAGKT captures information about complex knowledge structure and affects the knowledge tracking. Table 4 shows the experimental results of the

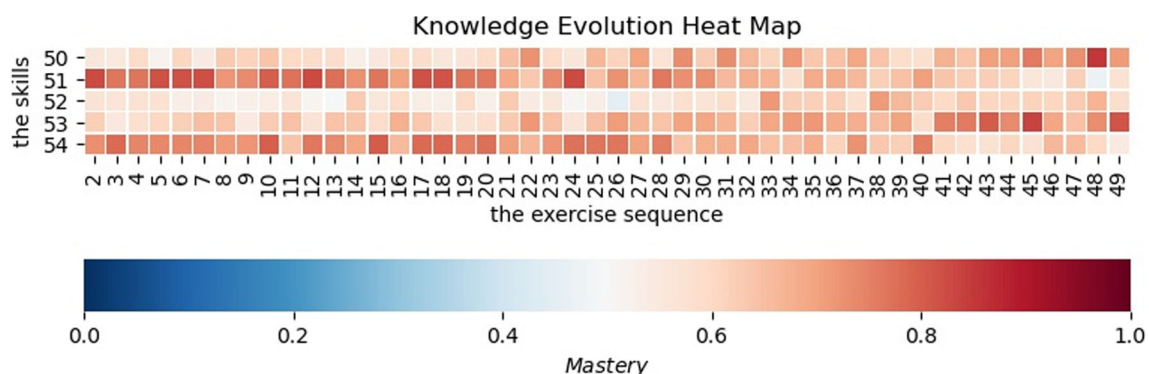
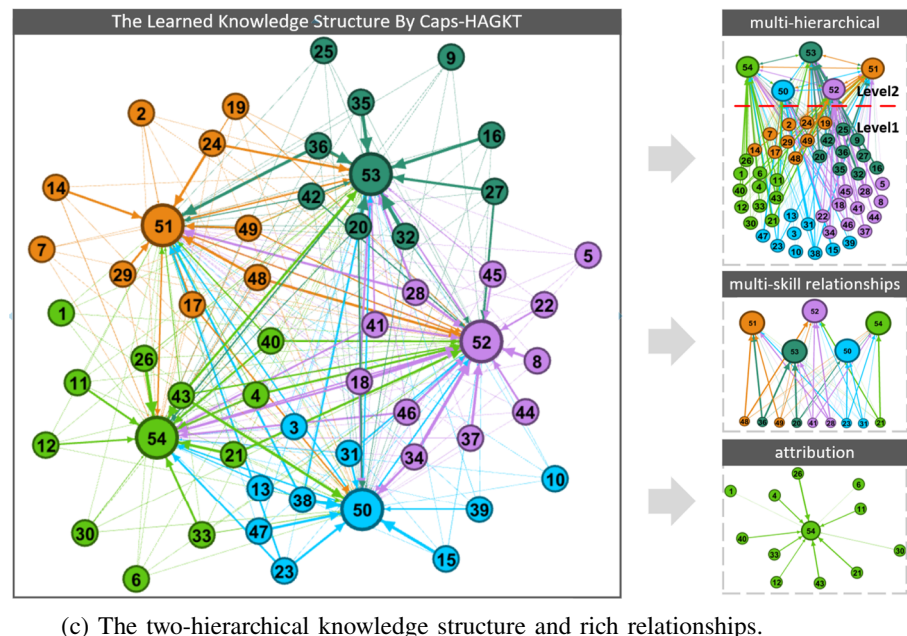
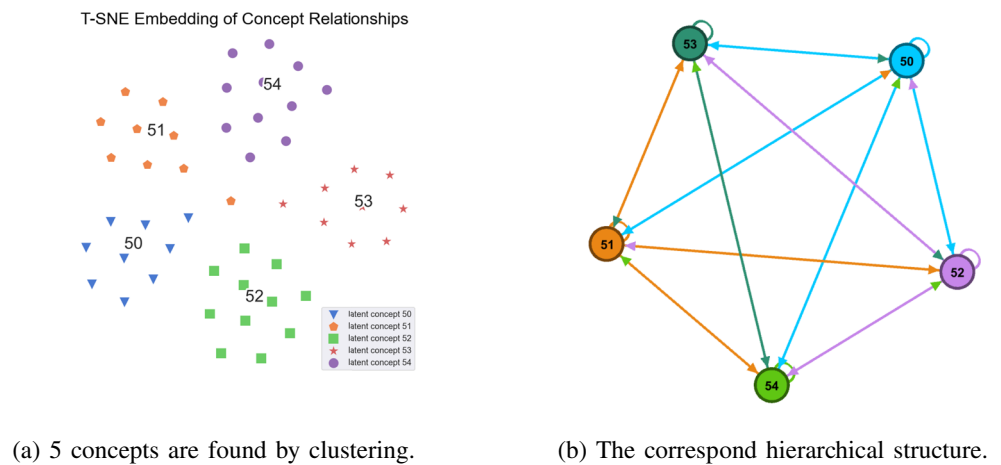


Fig. 7 An illustration of how a student's knowledge states are evolving for a sequence of 50 exercises in the dataset Synthetic-5

Fig. 8 Visualization of the latent knowledge graph of the model for dataset Synthetic-5, where colors of the nodes indicate the category of high-level concepts, the size of a node is proportional to its degree, and the width of a directed edge indicates its degree of dependence



default Caps-HAGKT structure and its various variants on all the datasets.

No meta-path: It is a variant in which only the semantic attention mechanism of knowledge capsules is deleted from the default structure, and it loses the ability to utilize different semantics. In the case of ASSIST2009 and Statics2011 datasets, the impact of deleting this layer is not as obvious as in the case of ASSIST2017 and ASSIST2015 datasets. This may be due to the fact that the first two datasets themselves are relatively simple, and the performance of learners is greatly affected by recent learning interactions.

No GNNs: It is used to automatically extract the structure of knowledge between the concepts at the same level (such as prerequisite dependencies). After “No meta-path”, we eliminated the graph network of knowledge capsules. In the case of ASSIST2017 dataset, on the contrary, it was observed that the predicted accuracy was improved. This

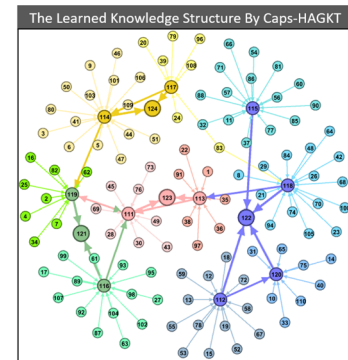
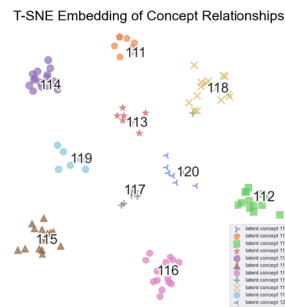
may occur as a result of fully automatic learning of the structure of latent knowledge.

No capsule: This elimination demonstrates the necessity of modeling different hierarchical knowledge structures (such as belonging dependence) and high-level concepts. The experimental results show that the knowledge tracking is severely affected on all the datasets.

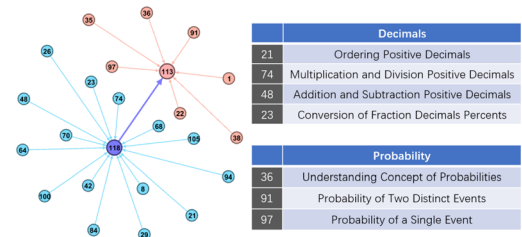
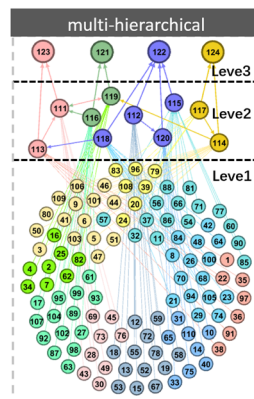
1 to 3 high-level capsule layers: These variants demonstrate the importance of multi-hierarchical high-level capsules. Increasing number of high-level capsule layers can achieve consistent improvement in predicting accuracy on the all data. It is worth noting that, in order to obtain interpretable predicted results, the high-level capsules need to be set according to the dataset and requirements.

Predefined graph: This variant injects the knowledge structure, predefined by experts, in the default model. The experimental results prove the important role of the knowledge structure between the concepts at the same level.

Fig. 9 Large scale knowledge graph of model learned in the real-world dataset ASSIST2009. The color of the node indicates the category of high-level concepts, the size of the node is proportional to the degree, and the width of the directed edge indicates the degree of dependence



(a) 10 concepts are found by clustering. (b) The corresponding implicit knowledge structure.



(c) The three-hierarchical graph structure. (d) Local connections with reasonable text tags.

7 Conclusions and future work

We proposed Caps-HAGKT can provide both high prediction accuracy and interpretable knowledge tracking results. We use multi-hierarchical knowledge capsules and graph network with external memory matrix to effectively capture complex knowledge structures (including multiple skills, prerequisites, attribution) with out any expert pre-definition, which are hidden behind learning resources.

We use knowledge capsules semantic attention to integrate multiple knowledge semantics and avoid the weakening of important knowledge features. In 5 datasets of online education platforms, our experiment works of baseline comparative and case analysis have proved the effectiveness of Caps-HAGKT.

In future, we will study: 1) Although we use text to help extract the latent knowledge structure, the course tags and hierarchical knowledge structure expressed in short

Table 4 Ablation study

Architecture	ASSIST2017	ASSIST2015	ASSIST2009	Synthetic-5	Statics2011
default	0.742	0.765	0.824	0.852	0.848
No meta-path	0.731	0.69	0.816	0.842	0.839
No GNNs	0.733	0.714	0.811	0.834	0.829
No capsules	0.527	0.515	0.614	0.631	0.569
1 high-level	0.729	0.669	0.780	0.812	0.809
2 high-levels	0.733	0.698	0.810	0.834	0.825
3 high-levels	0.741	0.736	0.791	0.838	0.827
Predefined graph	0.744	0.767	0.824	0.850	0.847

The bold marks indicate the best results

texts are still difficult to use. 2) When the expert's pre-definition is not used, automatic modeling of the complex knowledge structure of the knowledge capsule at same layer is insufficient, and the knowledge capsules graph network needs to be improved to better participate in the routing calculation. 3) Our model does not consider the important impact of forgetting behavior and contextual time interval on the prediction task and interpretability, and cannot evaluate the evolution of knowledge based on the time interval. 4) Need to optimize the routing strategy to reduce the amount of parameters and computational overhead.

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Zeyu He received the BE degree in computer science and technology from Beijing Information Science and Technology University (BISTU), China, in 2010. He is currently working toward the Ph.D. degree in the School of Chinese Academy of Sciences (CAS). His main research interests include graph neural network, recommender systems and intelligent education systems.



Wang Li received the BE degree in Communication Engineering from Shandong Normal University (SDU), China. She received the MA degree in the School of Chinese Academy of Sciences (CAS). Her main research interests include pronunciation assessment and acoustic model. He has published several papers in referred conference proceedings, such as Interspeech 2019.